

CAT Quantitative Aptitude & Speed-Math Shortcuts Handbook

Target Exam: CAT 2026 | Stream: Management | Subject: Quantitative Aptitude

Verified Official Academic Preparation Material - TheCampusNova Learning Portal

Key Topics Covered: Arithmetic, Algebra, Number Systems & Modern Math

1. Arithmetic: Percentages (fraction equivalents: $1/7 = 14.28\%$, $1/8 = 12.5\%$, $1/9 = 11.11\%$), Profit & Loss markup formulas
2. Time, Speed & Distance: Average Speed for equal distances = $2 \cdot S_1 \cdot S_2 / (S_1 + S_2)$ | Relative speed in same vs opposite direction
3. Algebra: AM $\geq$ GM inequality for positive reals | Quadratic roots nature ($D = b^2 - 4ac$) | Modulus equation graph methods
4. Geometry: Stewart Theorem, Apollonius Theorem, Inradius ($r = \text{Area} / s$), Circumradius ($R = a \cdot b \cdot c / 4 \cdot \text{Area}$)
5. Number Systems: Euler Totient function, Remainder Theorems (Wilson, Fermat Little Theorem, Chinese Remainder Theorem)
6. Exam Strategy: Target 10 to 12 correct questions with 100% accuracy in Quant to secure 99+ percentile; avoid negative guesses.